
MODEL SELECTION AND VALIDATION REPORT (remove_10_biomass model)

OBSERVATIONS:1651

SANITY: OK - all models in the selection process passed sanity checks.

SELECTED RANDOM TERM: spatiotemporal_random_fields_ar1

MODEL SUMMARY (remove_10_biomass model)

Spatiotemporal model fit by REML ['sdmTMB']

Formula: fixed_formula

Time column: year

Family: lognormal(link = 'log')

Conditional model:

	coef.est	coef.se
(Intercept)	4.93	0.18
remove_10_kde_fric	-0.35	0.04
remove_10_kde_feve	-0.46	0.03
sbt_mean	0.35	0.10
sbt_sd	-0.02	0.05
sst_sd	0.04	0.05
log_chla_mean	0.10	0.04
depth	-0.15	0.05
log_sum_fishing	0.01	0.04

Dispersion parameter: 1.12

Spatiotemporal AR1 correlation (rho): 0.89

Matérn range: 211.89

Spatiotemporal marginal AR1 SD: 0.87

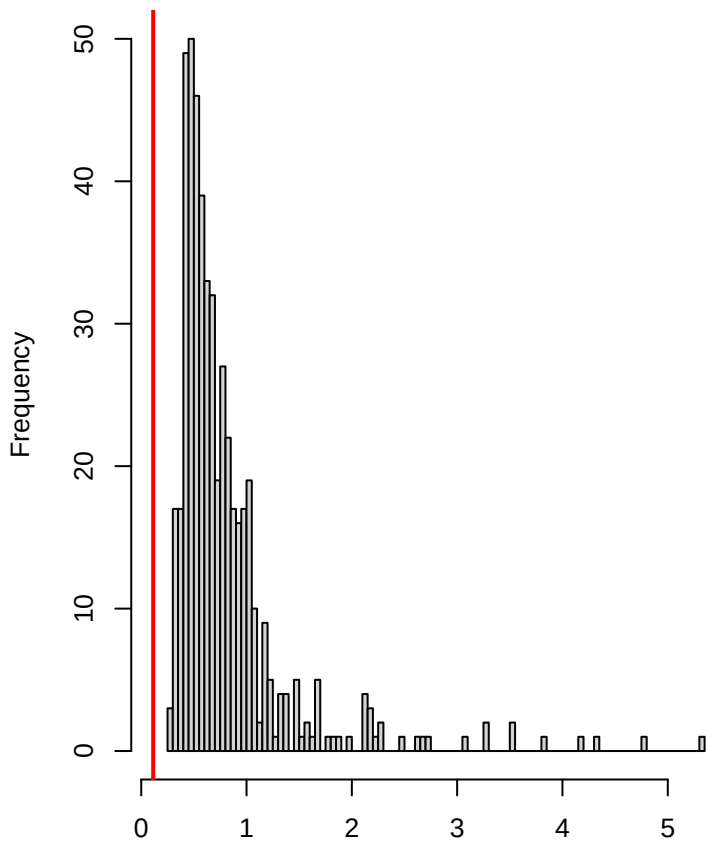
REML criterion at convergence: 9788.484

See ?tidy.sdmTMB to extract these values as a data frame.

SELECTION STATISTICS

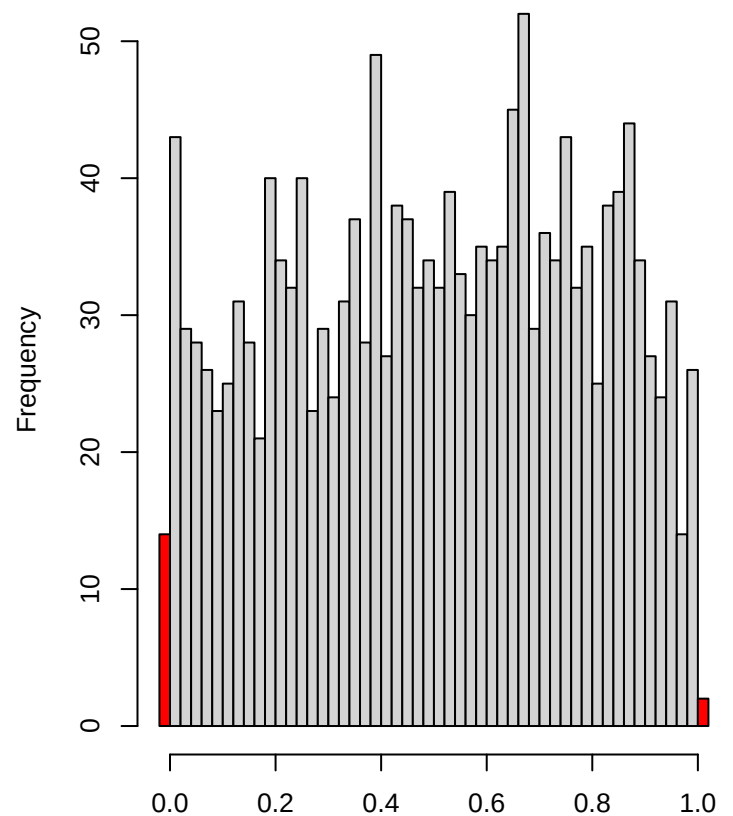
	df	AIC	delta_AIC
spatiotemporal_random_fields_ar1	13	19602.97	2.137446
spatiotemporal_random_fields_ar1_with_anisotropy	15	19605.11	96.241987
spatial_random_fields	12	19701.35	2.275613
spatial_random_fields_with_anisotropy	14	19703.62	34.217110
spatiotemporal_random_fields_iid	12	19737.84	3.468816
spatiotemporal_random_fields_iid_with_anisotropy	14	19741.31	137.430486
error_only	10	19878.74	NA

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated



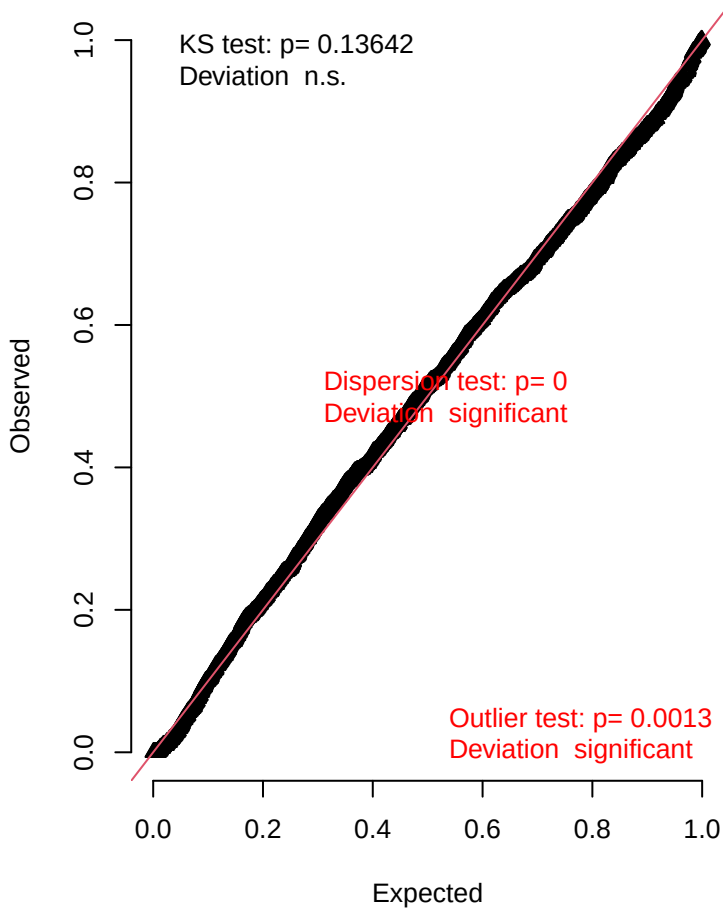
Simulated values, red line = fitted model. p-value (two.sided)

Outlier test significant

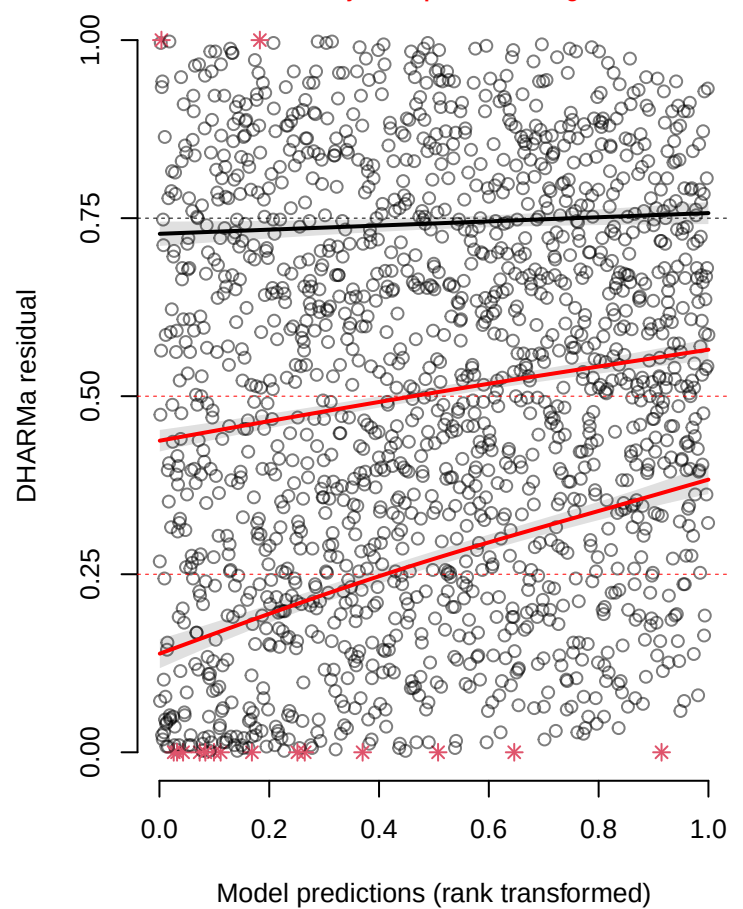


Residuals (outliers are marked red)

QQ plot residuals



DHARMa residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



DHARMA Moran's I test for distance-based autocorrelation
Moran's I = 0.032 | H1: Distance-based autocorrelation | p-value = 0.001

